Saptarshi Bagchi Full-Stack Developer | 3rd Year CSE(IoT) Student | React & Spring Boot Developer

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PROFESSIONAL SUMMARY

A 3rd-year B.Tech (CSE-IoT) student with hands-on experience building full-stack web applications, embedded systems integrations, and developer tools, backed by a strong foundation in DSA, OOPS, and REST API design.

EDUCATION

B.Tech — Computer Science & Engineering (IoT), 09/2024 – 06/2028
Institute of Technical Education and Research, SoA University Bhubaneswar
Relevant Coursework includes DSA, OOPS, FPGA Programming
CGPA : 8.19

Higher Secondary Education (Science, CBSE), *Sudhir Memorial Institute* 04/2022 – 04/2024
Completed 10+2 under CBSE board with Science stream Madhyamgram

TECHNICAL SKILLS

Languages: Java, HTML, CSS, TypeScript, JavaScript, Python

Frameworks & Libraries: React, Vite, Express, Next.js, Node.js, Spring Boot, Tailwind CSS, Framer Motion

Tools & Platforms: Git, GitHub, Maven, Docker, Postman, Vercel

Core Concepts: DSA, OOPS, REST APIs, FPGA Programming

Database: MongoDB, PostgreSQL

PROFESSIONAL EXPERIENCE

Elevance Skills, *Full Stack Developer Intern* 06/2026 – 07/2026
(Java, Maven, Spring Boot, MongoDB, Next.js, TypeScript, Tailwind, Docker, Render)
(https://make-my-trip-clone-a9q3.onrender.com)

- Built a Spring Boot REST backend with 22 endpoints across 5 controllers and 3 MongoDB collections embedding bookings, refunds, and reviews, plus BCrypt password hashing and centralized CORS.
- Built a responsive Next.js frontend with shadcn/ui and Tailwind CSS, featuring 5 routes, 17 custom components such as a seat map and price-history chart, and 20 Axios calls to the Spring Boot backend.
- Containerized the Spring Boot backend with a multi-stage Docker build on Java 21, running as a non-root user to reduce the container's attack surface, and deployed it on Render.

PROJECTS

Posture Correction Desktop App, *(Electron, React, Python, pySerial, ESP32, MediaPipe, Gemini API)*
(https://posture-identifier-application.vercel.app)

- Built a desktop app streaming ESP32 spine-alignment data via a Python serial reader and Electron to a React UI.
- Designed corrective feedback through native desktop notifications, break reminders and an on-screen overlay.
- Integrated MediaPipe and Gemini API to measure deviation angles and generate posture correction plans.

Mosquito Risk Dashboard, *(React, JavaScript, React Router, Framer Motion, OpenStreetMap)*
(https://mosquito-risk-detection.vercel.app)

- Implemented a React dashboard that maps mosquito-risk zones and shows live metric cards for at-a-glance monitoring.
- Visualized risk patterns with a 365-day activity heatmap and a 7-day forecast view.
- Added an OpenStreetMap location view and client-side image-upload previews.

NYC Wordle Game, *(React, Vite, JavaScript, Python)*
(https://wordle-simulator-fawn.vercel.app)

- Developed a browser-based Wordle clone replicating core gameplay mechanics using React and Vite.
- Implemented keyboard input handling and tile management to support dynamic, row-by-row word entry.
- Generated a 5,000-word answer list with Python's wordfreq library, selecting a random target word each round.

ACCOMPLISHMENTS

- Maintained a 250+ day LeetCode streak, practicing core DSA topics and Dynamic Programming.
- Built a Sensor-Data Dashboard for an IoT project, placing Top 50 out of 300+ teams at SIH Ideathon 2026.

Certificates

Introduction to PostgreSQL — Great Learning (Certificate ID:JFCZYAMP)